Structural Equation Modeling with Onyx

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Preface

This is a self-paced workshop for Onyx, a graphical user-interface for structural equation modeling.

![images/somemodels.png]

We assume that you already familiar with the basics of Structural Equation Modeling.

1 Introduction

Onyx's interface is somewhat unusual as it presents the user with an empty desktop. All major interactions in Onyx are via context-menus or left- and right-mouseclicks that provide context-dependent shortcuts.

This workshop is work-in-progress and has several chapters targeted at readers that are familiar with Structural Equation Modeling and want to learn about how to best use Onyx in their scientific work.

There is no need to follow the order of the modules as they are presented. Pick a topic about which you would like to learn more. The topics contain some basic information, some demonstrations, and practical exercises for you to follow along.

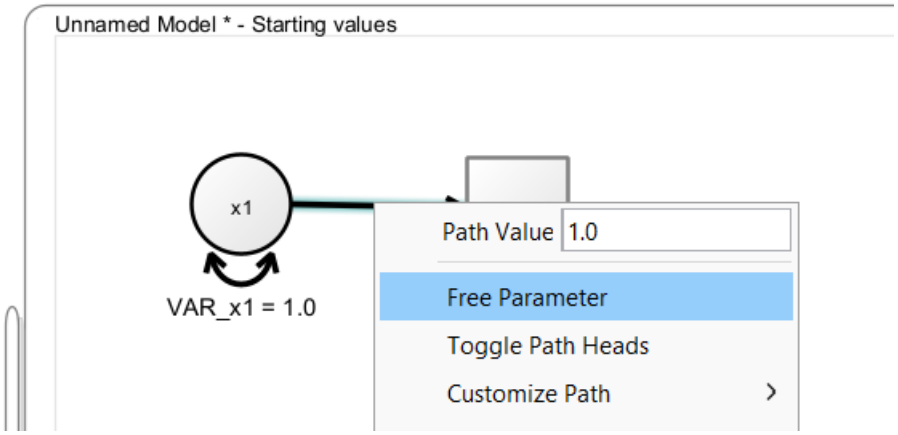
2 Basic Modeling

To create a model, either right-click on the empty desktop and select “Create new Model -> Create Empty Model” or double-click on the desktop. This creates an empty model view

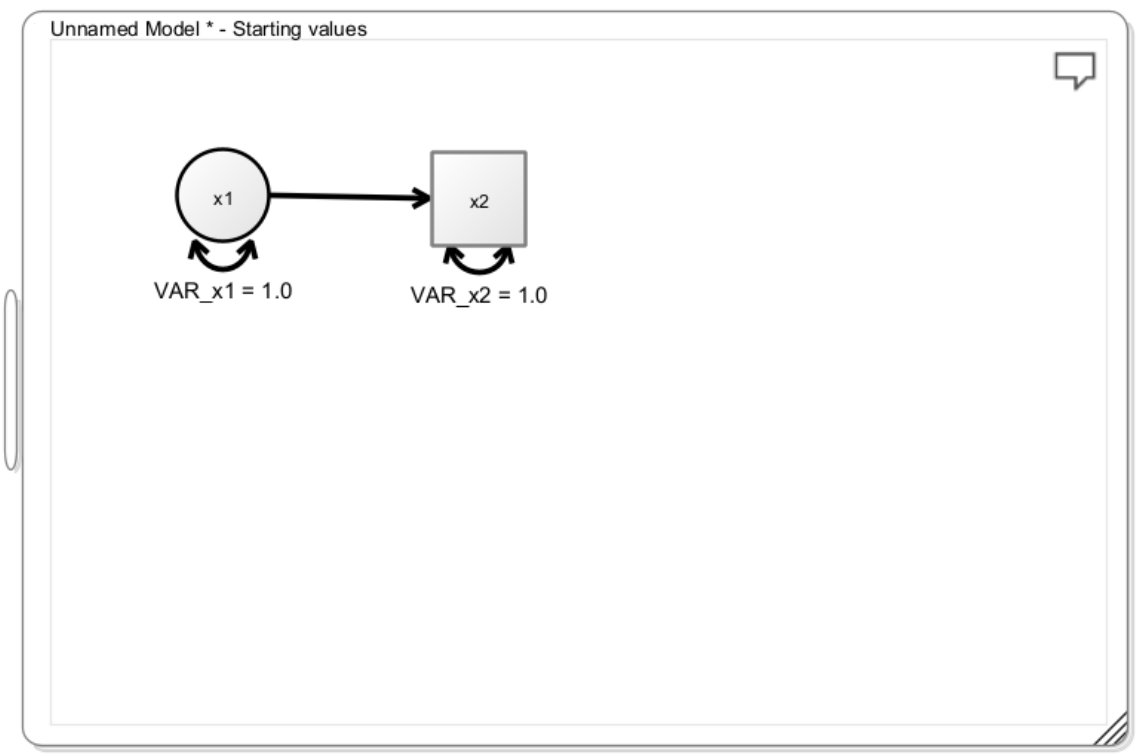
You can now add variables, by either right-clicking on the empty view and selecting “Create Variable”. Then you can either add a latent variable (depicted as a circle), an observed variable (depicted as a rectangle), a constant variable (depicted as a triangle). Beyond standard RAM notation , you can also add a multiplication node that allows you to model latent interactions. As a short-cut, you can simply double-click on an empty space in the model, to create a latent variable or double-click while holding down the SHIFT-key to add an observed variable.

To add a path between two variables, press the right mouse button on a variable, hold it down while dragging the path on to the target variable, then release the mouse button. This creates a regression path with a value fixed to “1” between the variables. When you hold down SHIFT

while doing so, you create a covariance path. Right-click the path and select “Free Parameter” if you want this path to be freely estimated.



Try to add a latent and an observed variable and connect them with a path from the latent to the observed variable, and your model should look like this:

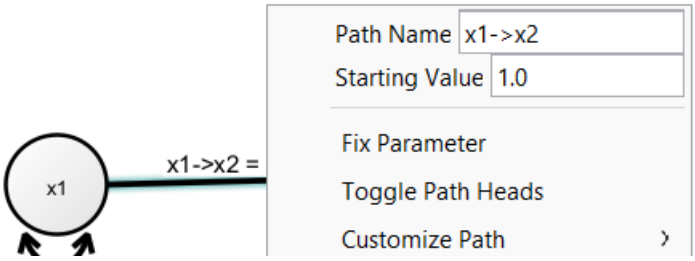


2.1 Changing Visuals

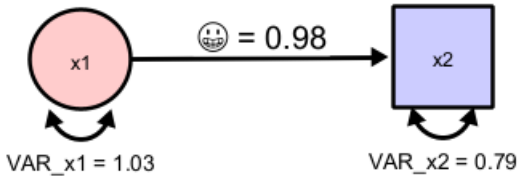
Onyx has a variety of options to adjust the appearance of the graph. Right-click on a node and choose "Customize Variable" to explore various options to adjust font size, line colors, fill colors, etc. The same is possible if you choose "Customize Path" when you right-click on a path.

2.2 Renaming Parameters

If you free a parameter on a path (representing either a variance, covariance, a regression, or a mean), you can change the name of the parameter by right-clicking on the path and then changing the "Path Name". By default, these parameters are named based on the adjacent variables but you can choose your own name here.



Note that Onyx has Unicode support, that is, you can directly use unicode characters such as letters from the greek alphabet. Thus, it would even be possible to use emojis as parameter labels even though I do not recommend that:

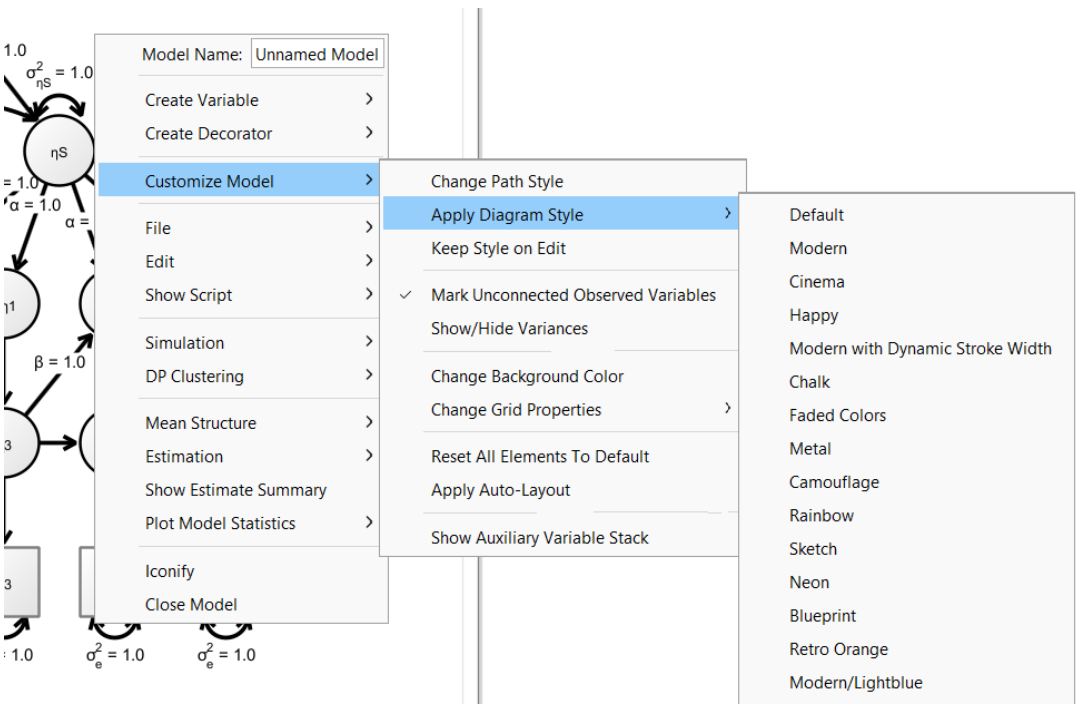


Onyx further supports a very limited subset of LaTeX commands to render parameter labels. This pseudo LaTeX input is triggered by starting labels with a dollar sign. This pseudo input supports lowercase and uppercase greek letters, as well as subscript and superscripts. Here are some examples that are valid parameter names in Onyx:

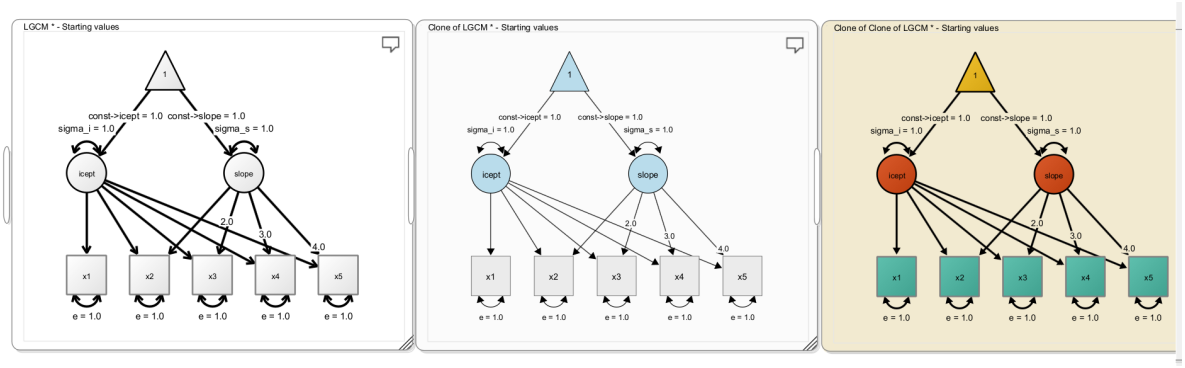
Input	Output
	α
^2	β^2
_{person}	γ_{person}
^2_i	Ω_i^2

2.3 Diagram Styles

Onyx ships some pre-made styles for your graphs. Choose "Customize Model -> Apply Diagram Style" and then choose from the available choices. Certainly, not all of them are useful for scientific communication but you may want to start with Modern or Modern/Lightblue. Further, you may want to choose "Keep Style on Edit", which continues to automatically apply the chosen style to newly added variables or paths in the diagram (otherwise, they are added using the default style).

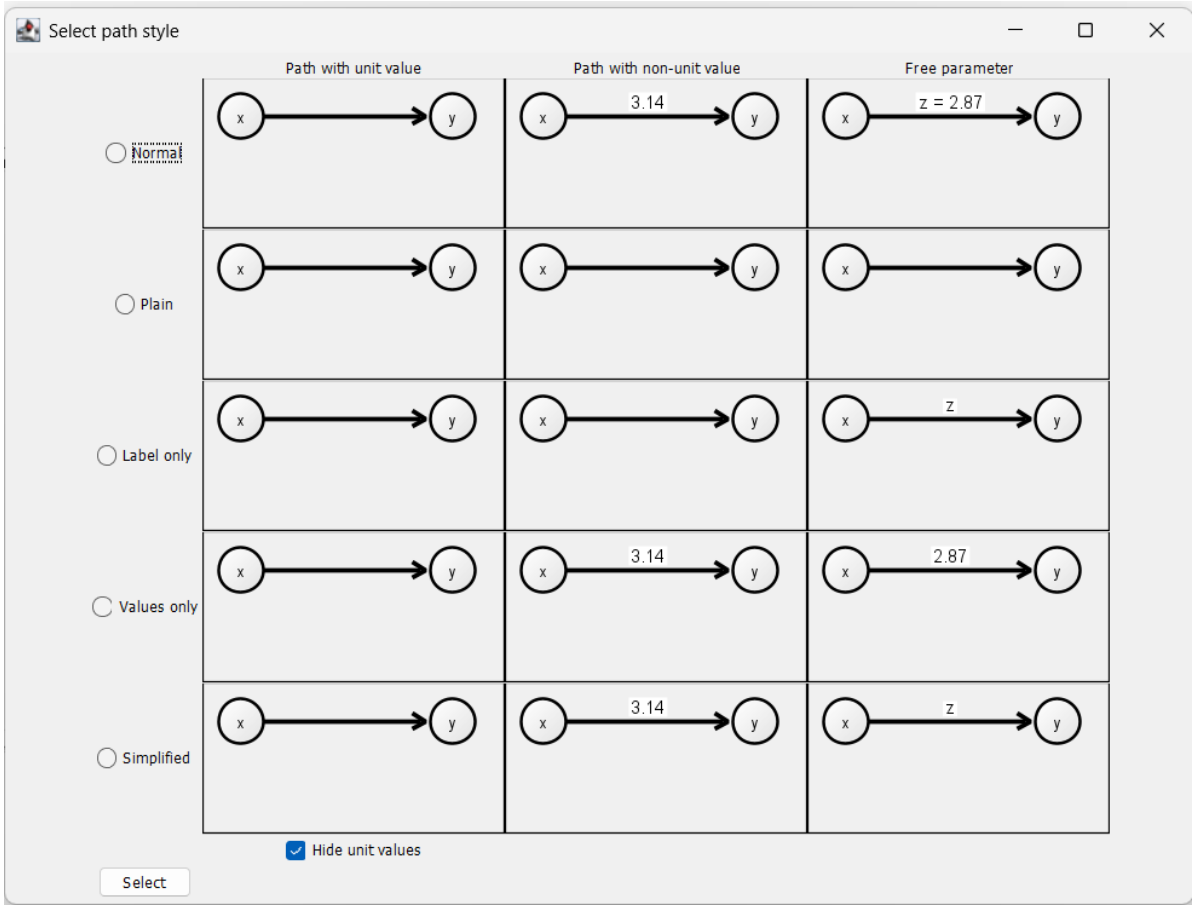


Here are three examples of styles:



2.4 Path Styles

By default, Onyx shows path with a value fixed to one without any label. Paths that have a value fixed but different than one are shown with the value as label. Freely estimated parameters are shown with both their label and their value (e.g., "z = 2.87"). You can also choose different path styles by right-clicking on a model, then selecting "Customize Model -> Change Path Style". This opens the following window in which you can select among several alternative options. In particular, the option "Simplified" may be useful to create figures for manuscripts.



2.5 Exercises

- Create a factor model with one latent factors, four observed variables, residual variances for each factor, a latent factor variance, and factor loadings; fix the first factor loading at "1" and let the others be freely estimated; also freely estimate all residual variances; rename the observed variables to "item1", "item2", ...; rename the factor to "factor", rename the factor loadings to lambda1, lambda2, ... but use greek letters.
- Apply diagram styles and make your own modifications
- Export the diagram as image file

3 Untitled

3.1 Model Fit

In SEM, the fit function to obtain maximum likelihood estimates under multivariate normality of the observations and assuming no missing data is:

$$F_{\mathrm{ML}} = \log|\Sigma(\theta)| + \mathrm{tr}\left(\mathbf{S}\,\Sigma(\theta)^{-1}\right) - \log|\mathbf{S}| - p$$

Assuming H_0 that there is no misfit is true, the following test statistic has an asymptotic χ^2 distribution

$$\chi^2 = (N-1)\,F_{\mathrm{ML}}$$

3.2 Model Fit

- χ^2 is the fundamental model fit measure as discrepancy between model-implied (co)variances and means and observed (co)variances and means
- Significance test of close fit is usually not recommended as it has very high power; instead we usually favor effect-size-type metrics
- CFI: discrepancy between an independence model and the hypothesized model; *CFI asks: How much better does my model fit compared to a model that assumes no relationships at all?*
- RMSEA: estimates how much misfit per degree of freedom your model has in the population. *RMSEA asks: "Is the model achieving good fit without overusing parameters?*

3.3 Residuals

Residuals represent the difference between what the model predicts and what is actually observed in the data. More specifically, SEM works by reproducing the observed covariance (or correlation) matrix using a model-implied covariance matrix. Residuals are simply the element-wise differences between these two matrices. If the model fits well, these residuals should be small—ideally close to zero—because the model is successfully capturing the relationships among variables.

Residuals are useful because they provide localized information about model misfit. Global fit indices like chi-square, CFI, or RMSEA tell you whether the model fits overall, but they don't tell you where it fails. Residuals, in contrast, point to specific pairs of variables where the model does a poor job reproducing the observed relationship. For example, if two indicators have a large positive residual, it means their observed correlation is higher than what the model predicts, suggesting that something is missing, such as a residual correlation or an additional factor.

3.4 Raw and Standarized Residuals

There are different types of residuals in SEM. Raw residuals are simple differences between observed and implied covariances, while standardized residuals scale these differences by their expected variability, making them easier to interpret across different variable pairs. Standardized residuals are often preferred because they help identify which discrepancies are unusually large relative to sampling variability.

Importantly, residuals are not just diagnostic tools – they also guide model refinement. Large residuals can suggest theoretically meaningful modifications, such as allowing two indicators to correlate due to shared wording or content (e.g., shared method variance). However, they must be interpreted carefully: adding parameters purely to eliminate residuals can lead to overfitting and models that do not generalize well.

3.5 Model-implied and observed covariances

A summary measure of the misfit in the residuals is the standardized root mean square residuals, which is:

$$\text{SRMR} = \sqrt{\frac{2}{p(p+1)} \sum_{i \leq j} \left(r_{ij} - \hat{r}_{ij}\right)^2}$$

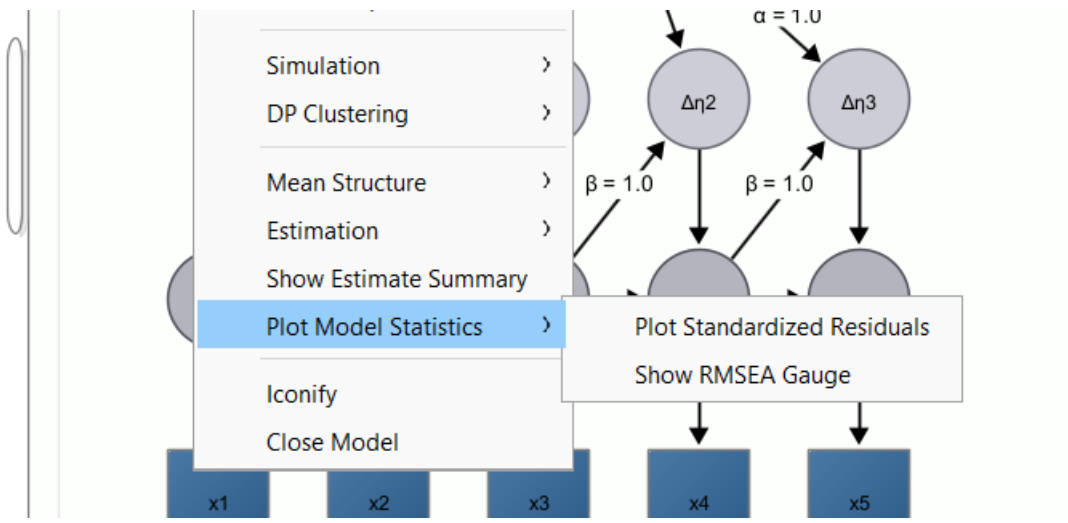
where

r_{ij} are the observed correlations $\hat{r_{ij}}$ are the model-implied correlations

The exact computation differs across programs. Some programs standardize such that the means are included and some don't. Some programs standardize such that the diagonal elements are included and some don't. Note that Onyx standardizes the observed covariance matrix and then the model-implied covariance matrix each with their own variances. Thus, Onyx SRMR does not include misfit in the means or the residual variances – just the covariances.

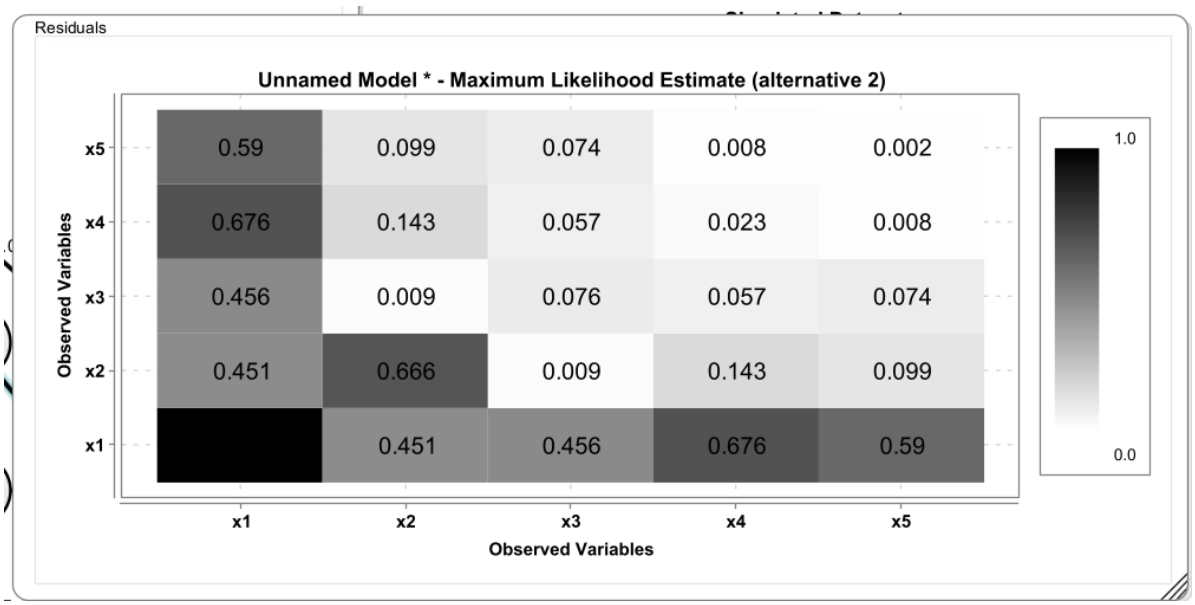
3.6 Model Fit Plots

There are two new plotting features to obtain visual information about model fit. Right-click on a model and select "Plot model statistics". Then choose "Plot Standardized Residuals"



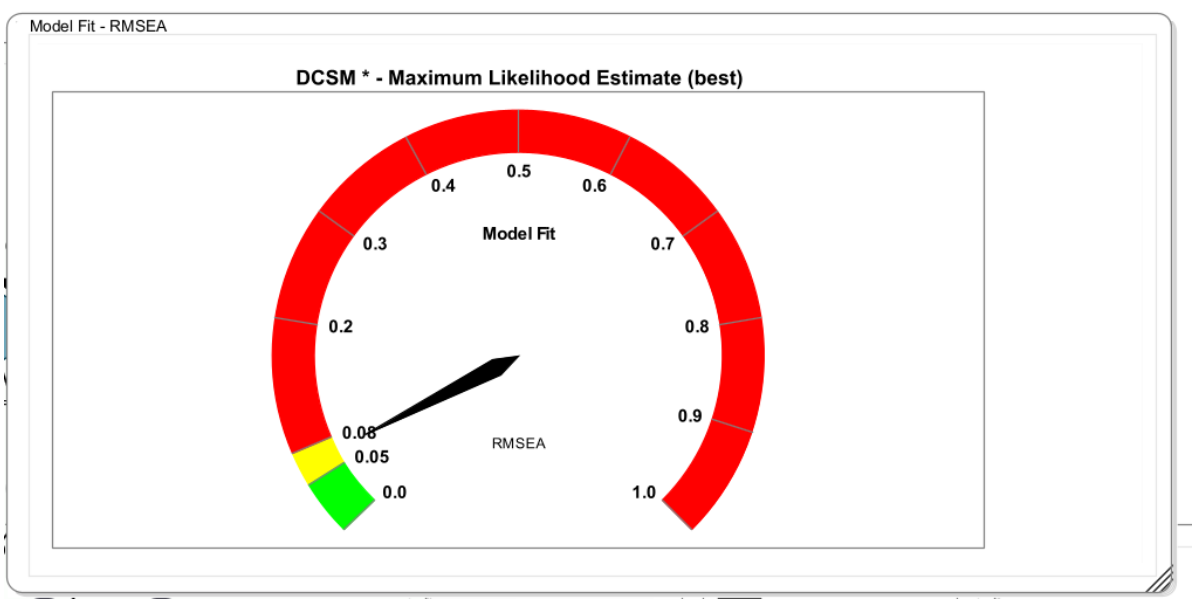
3.7 Residual Plots

This opens the standardized residuals plot. This plot show standardized residuals that are slightly different than what is used to compute the SRMR. These standardized residuals are computed such that they also represent misfit in the diagonal of the matrix, that is, the residual variances. This is achieved by standardizing both model-implied covariances and observed covariances with the variances of the observed covariance matrix.



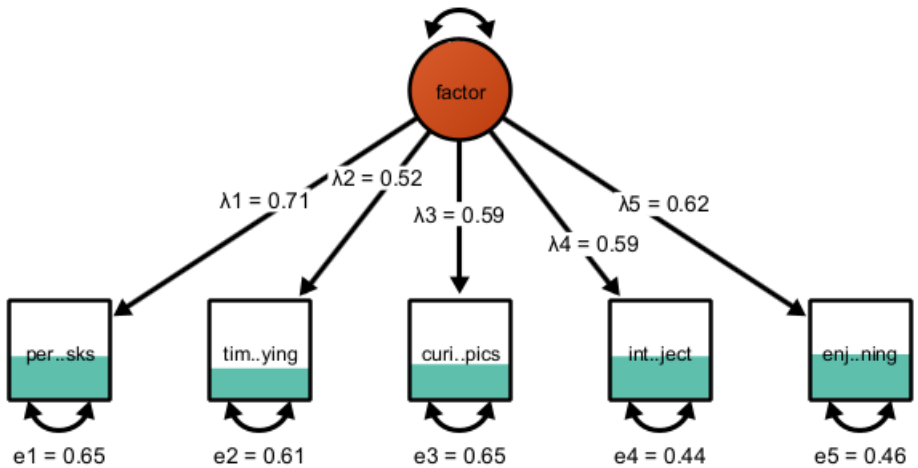
3.8 Model Fit Gauge

A further visualization of model misfit is the RMSEA gauge. This is a gauge that shows the RMSEA and shades some region of the RMSEA according to whether those ranges were proposed as ranges of acceptable or good fit.



3.9 Visualizing Explained Variance

For observed variables, you can select a fill-style that reflects the percentage of explained variance in a variable. Right-click on a variable or a selection of variables, click "Customize Variable", then "Select Fill Style", then "Bokers R2". This applies a fill-style that fills the selected variable only up to the level that corresponds to the variance explained by other variables in the model. The unexplained part is not filled. This gives a good quick overview, for example in factor models, how well an item works. Here is an example in a factor model with five indicators:

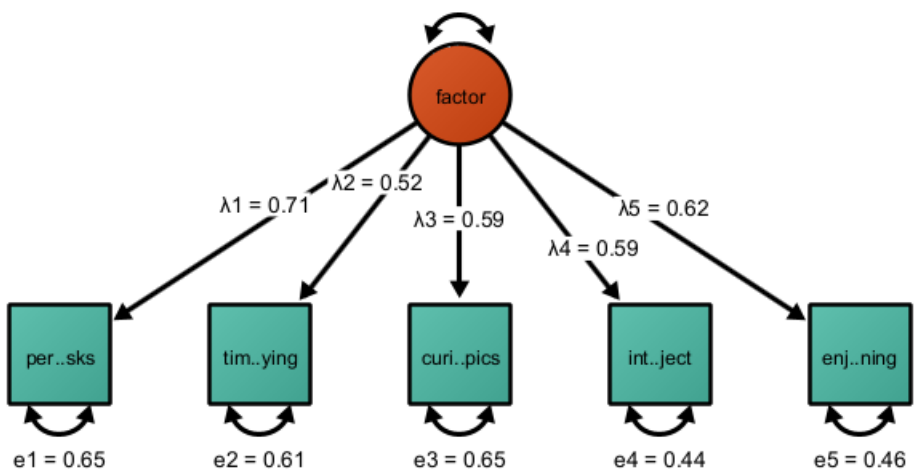


3.10 Exercise

- Open the dataset "academicmotivation.csv" in Onyx
- This is a simulated dataset of a common factor of academic motivation that is measured with five indicators: persistence_on_tasks, time_spent_studying, curiosity_about_topics, interest_in_subject, and enjoyment_of_learning
- Fit a common factor model (just covariances; no means); apply Boker's R2 style to the observed variables
- Inspect model fit, model-implied covariances, and residuals (using the plot)
- Given these statistics and the information that interest_in_subject and and enjoyment_of_learningtion tap more into intrinsic_motivation than the others, how could we modify the model?

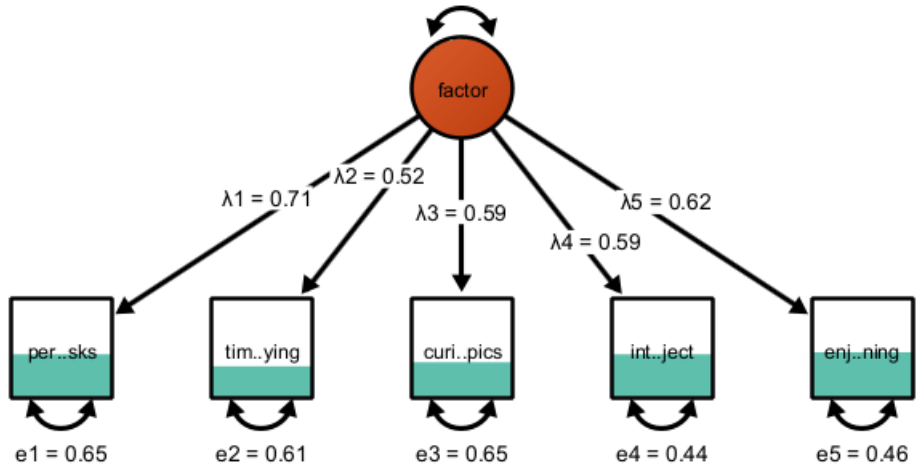
4 Solution

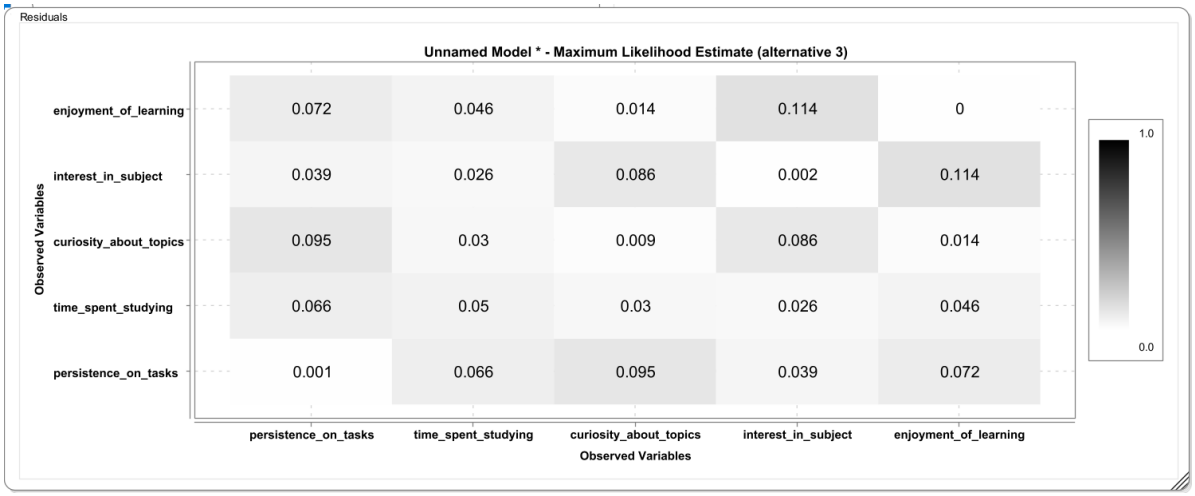
4.1 Factor Model



Model fit is $\chi^2 = 21.033$, RMSEA=0.106, CFI=0.106, SRMR=0.055. RMSEA and CFI are outside regions that are usually deemed acceptable.

Applying Boker's R2 style, this yiels:





The large discrepancy in the correlation of enjoyment_of_learning and interest_in_subject may be an indication that these items also capture something different than the other items. In light of a theory (intrinsic vs academic motivation), this can be used to modify the factor model to include a residual correlation (which is akin to a second common source of variance) to the model. By adding another free parameter, model misfit in the present sample is reduced. Note that such model modifications after having seen the data should always be reported as exploratory.

5 Simulate and Plot

> Goal: Learn about simulation and plotting of data

5.1 Simulation to Understand Model Parameters

We want to answer this question:

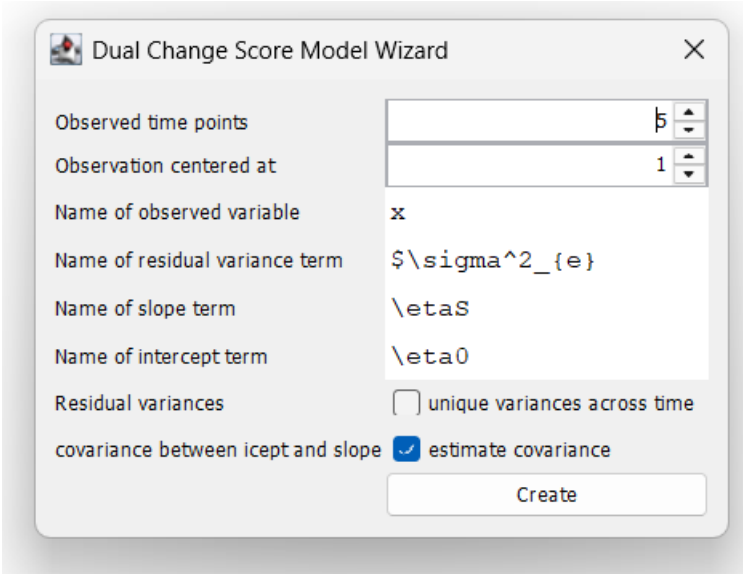
In a Dual Change Score Model, what effects do parameters have on the model-implied trajectories?

Steps:

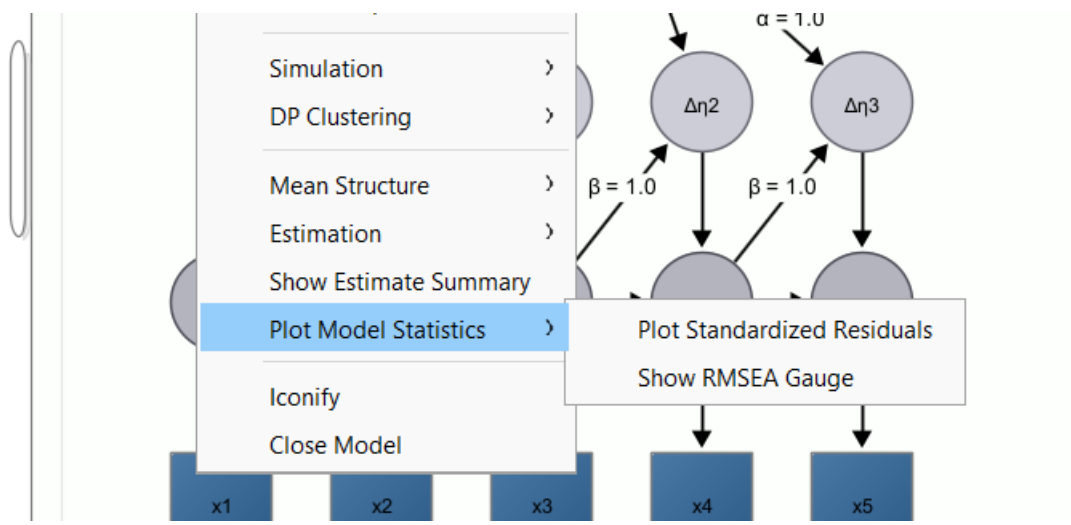
1. Create a Dual Change Score Model
2. Simulate Longitudinal Data
3. Plot Longitudinal Data
4. Change Model (Population) Parameters and Repeat from 2

5.2 Dual Change Score Model

Use the DCSM-Wizard to create a dual change score model. Keep all values at their defaults. This creates a model with five time points and the observed time points will be called x1, x2, ..., x5.

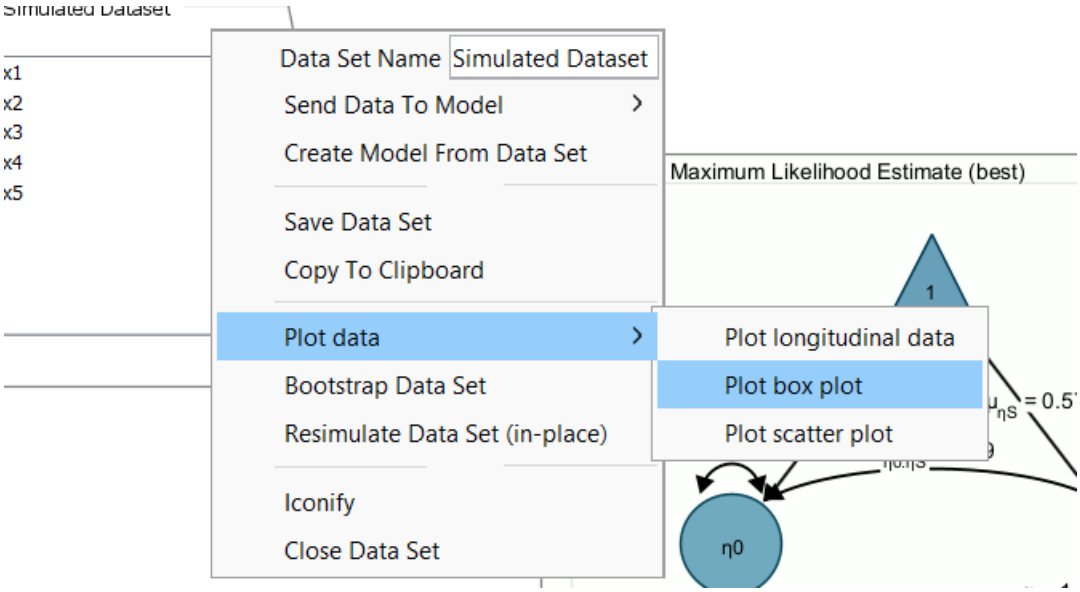


5.3 Simulation

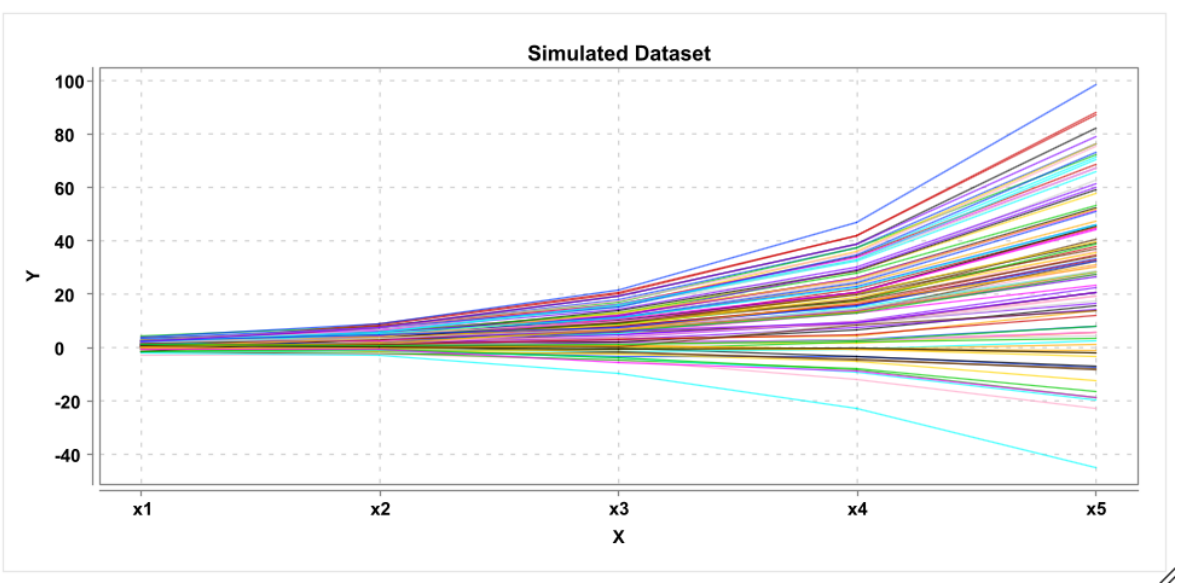


5.4 Plot

Next, right-click on the simulated dataset, choose "Plot data" and select "Plot longitudinal data".

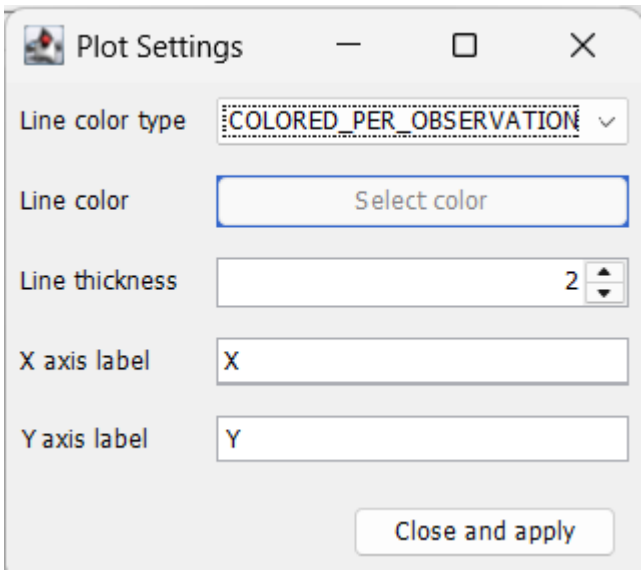


5.5 Longitudinal Plot



5.6 Customize the Longitudinal Plot

Right-click on the longitudinal plot window. This will show some basic plot settings. Try out different ways to customize the plot. By default, the line color is selected as "COLORED_PER_OBSERVATION" such that each observation gets a different color. Alternatively, you can set the color to "USER DEFINED" and then pick a line color. You can also change the line thickness, and the axes labels



Click "close and apply" to apply the settings and close the dialog window. Right-click again and choose "save PNG" to save the graph as image file.

5.7 Plot subsets of variables

If no variables are selected in the dataset, the longitudinal plot will be plotted across all variables in the dataset (in the order, in which they appear in the dataset). If you want to plot only across a subset of variables, select those variables in the variable list (e.g., by holding CTRL or CMD down and select the respective variables) and then click "Plot Data -> Plot longitudinal data".

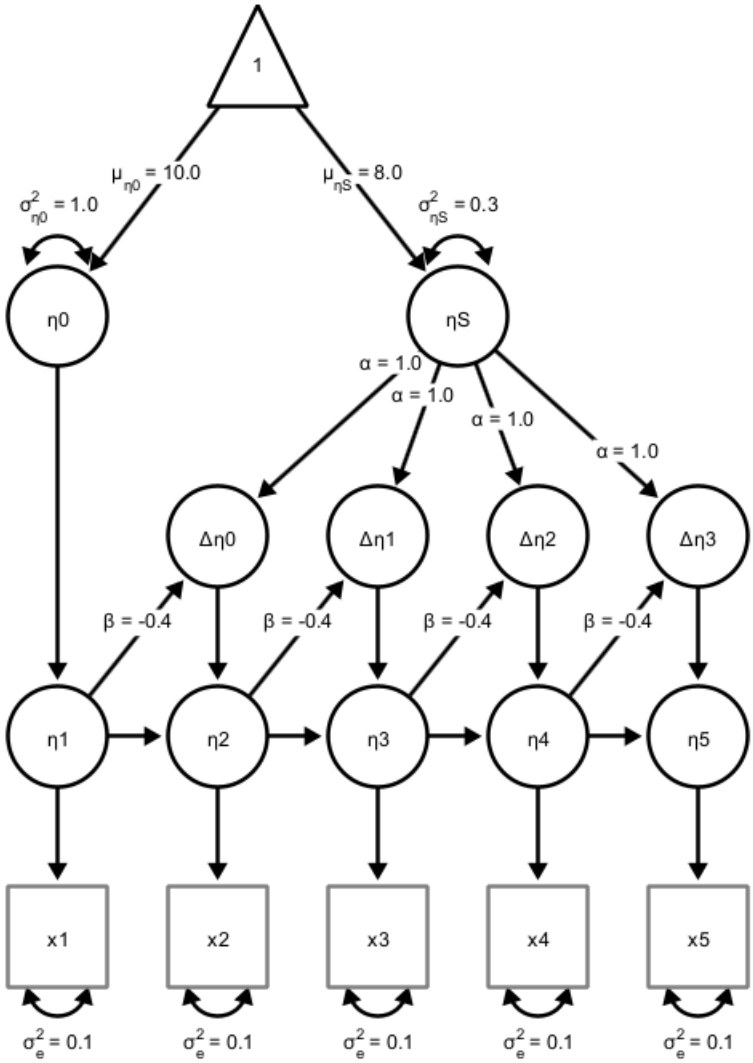
5.8 Exercises

- Create a dual change score model using the wizard
- Simulate Data and plot the data (longitudinal plot)

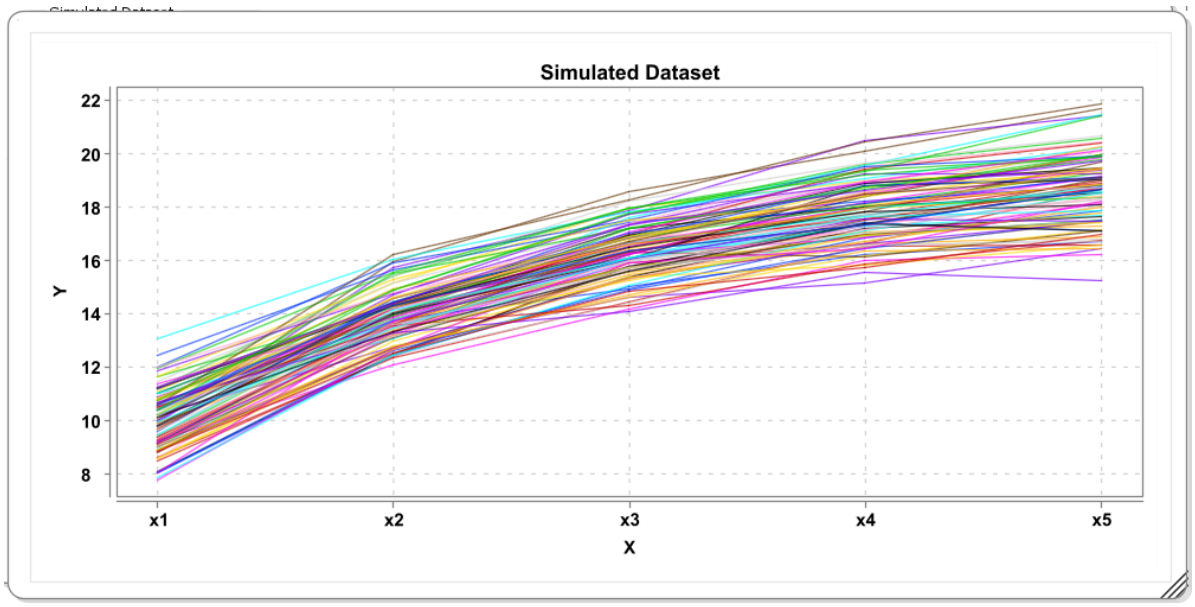
- Change Parameters of the model, resimulate and plot
- What happens if the self-feedback (beta) is 1, 0, or -1? Why?
- If you set the self-feedback to zero, what are the effects of changing alpha? What if you set different alphas (rename them to alpha1, alpha2,… first) for different time points?
- By setting a form of constant positive change (alpha) and negative proportional change (beta), can you generate simulated learning curves that reach an asymptote?

6 Solution

The self-feedback term beta implies a form of proportional growth if values are positive because change then depends on previous level. This by itself can reflect an exponential growth function. In combination with constant growth rates (alpha > 0), negative proportional change can be used to model change that flattens after a while. Here is one possible solution:



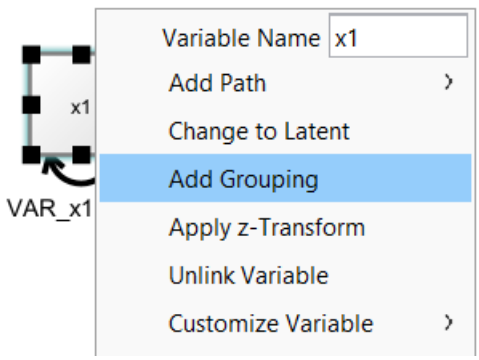
And these are some simulated trajectories from the model (N=100):


Simulated Dataset
22
20
18
16
14
12
10
8
Y
x1
x2
x3
x4
x5
X

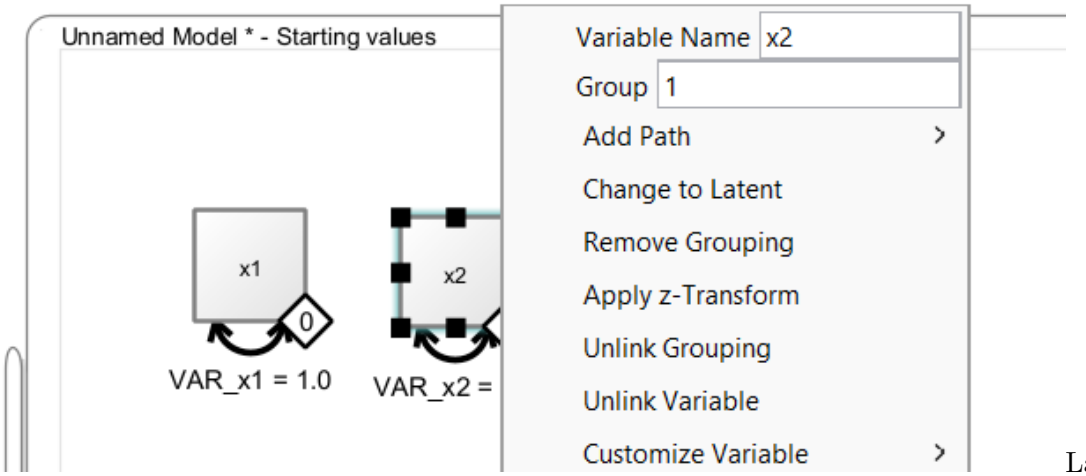
7 Multi-Group Models

Onyx support multi-group models, in which we use a grouping variable that encodes group membership between individuals. Onyx has its own approach to multigroup modeling which works with filters on observed variables that filter out subsets of observations that should be modeled by a given observed variable.

By right-clicking on an observed variable, you can select "Add Grouping" to add a filter to one or more selected observed variables.



Then, a little diamond grouping indicator appears on that variable. The value inside the diamond reflects the group membership. To change group membership, right-click the diamong and change the "Group" value:



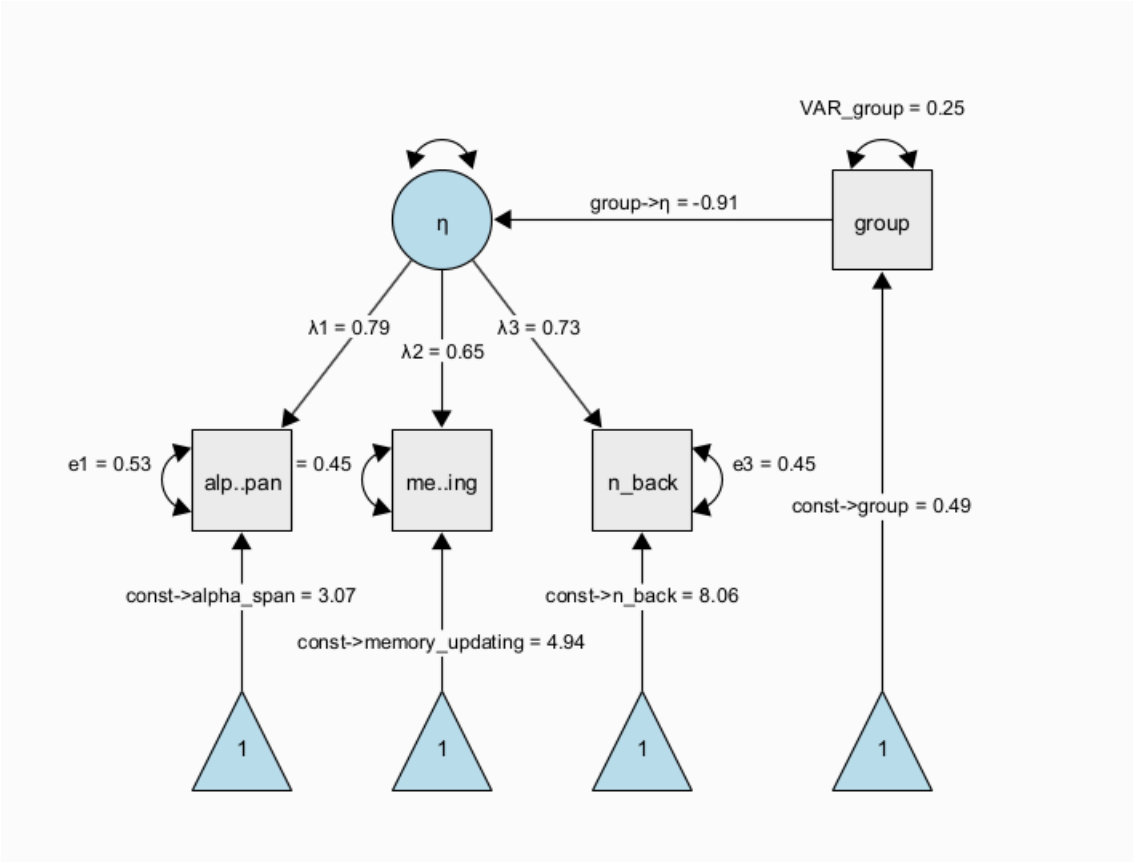
Latent Mean Differences

The diamonds are variable containers, that is, you can drag a discrete-valued grouping variable from a dataset on to the diamond. As long as the diamond is not filled (white fill color), it is not connected. Once it is connected, it becomes filled (black color), that is, it is now associated with a grouping variable in the dataset.

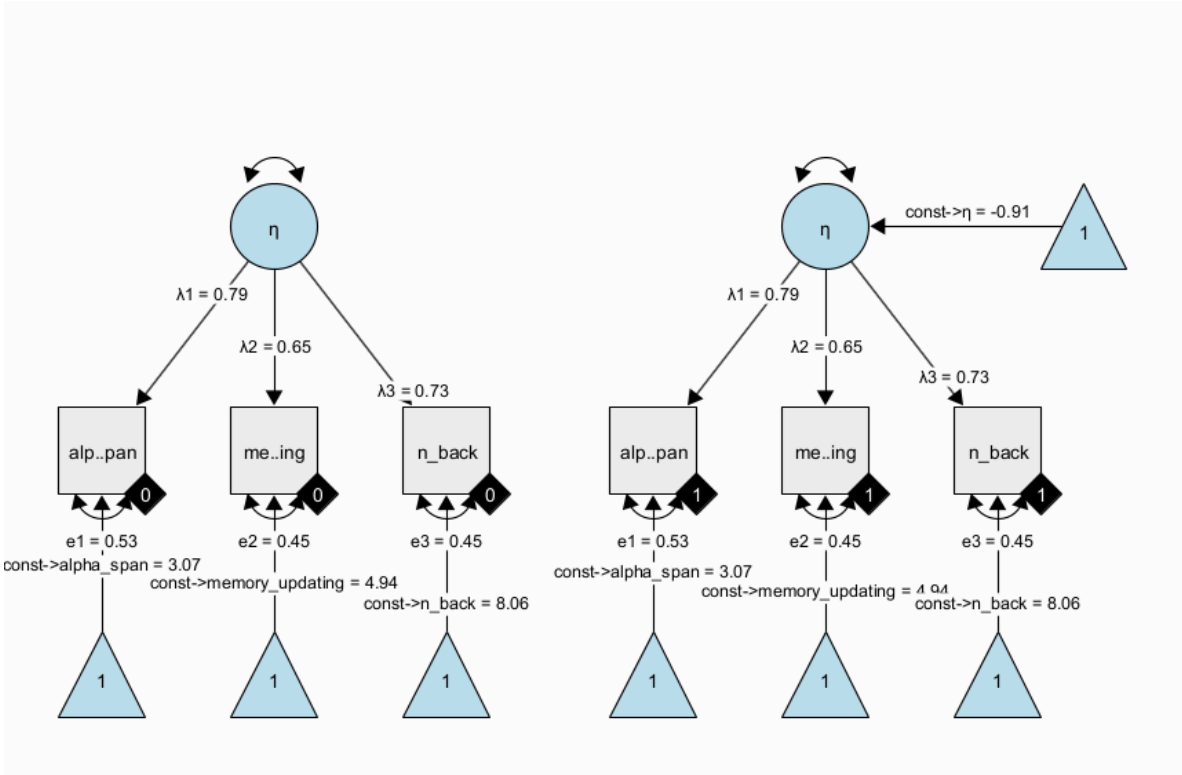
8 Latent Means

Models of latent mean differences are like the classical t-test or ANOVA but estimated on latent constructs.

Figure 1 shows a path diagram representing one possibility to set up a latent mean difference model when there are multiple, noisy measurements of the same underlying latent construct. This model is a MIMIC (multiple-indicators multiple-causes, **hancock2001effect?**) model, in which we estimate a measurement model of a latent trait η across two groups that are encoded via a dummy group indicator. The latent trait is measured through multiple observed indicators (here, $x1$ to $x3$) with indicator-specific residual variance terms, which allows to separate error variance from true-score variance. If correctly specified, this approach allows us to obtain more precise estimates of the latent trait purged of measurement error and unreliability (DeShon 1998). In this specific instance, the regression path from the dummy-coded group indicator to the latent variable directly represents a standardized mean difference because the scale of the latent variable was fixed to have a variance of one. Note that the MIMIC model assumes that measurement invariance holds, that is, the factor loadings, item intercepts, and item residual variances are assumed to be the same across groups.



A more flexible model specification is also possible; using a multiple-group factor model, we can estimate the same factor model in each group separately and empirically test whether measurement invariance holds; then, we can continue to test whether there is a difference in latent factor means across groups. This is also known as the SMM [structured means modeling; (**hancock2001effect?**)] approach:



Note that all parameters with the same name have identity constraints; in particular, this means that parameters with the same name are identical across groups. In the path diagram shown above, all parameters are identical across groups. This implements a very strong assumption of measurement invariance across groups and identity of magnitude of variance (= individual differences at the latent level) across groups. Again, this restricted version of the SMM model is identical to the above dummy-coded MIMIC approach. However, the SMM allows to also freely estimate parameters between groups and perform tests of whether some of the restrictions are supported by the data or not.

8.1 Exercises

- Load the dataset "factor_model_memory.csv"
- Use the "Latent Means Model" wizard to create both the SMM and the MIMIC approach model to estimate models of latent mean differences
- Freely estimate the latent variance in the second group of the SMM approach
- Copy the SMM model and freely estimate the factor loadings across groups; run a likelihood ratio test between the models as a test of measurement invariance

9 Exercises

In the following chapters, you will find exercises to practice your Onyx and modeling skills.

10 Exercise: Multiple determinants of lifespan memory differences

In this exercise, we will reproduce results reported by (Henson et al. 2016). In their study, they speculate on different theories of memory. Here, we use SEM to test different measurement models representing the different theories of memory. Items are 9 memory scores (3 valences and 3 tests, fully crossed). The tests tap into associative memory, item memory, and visual priming. Valences are positive, neutral, negative. The data is based on the reported covariances of the item d' values across the nine tests. Henson et al. (2016) write in their introduction:

> Human memory is often claimed to consist of different systems, supported by distinct brain regions[1],[2],[3]. However, the precise fractionation of these memory systems is a matter of continuing debate[4],[5].

In the following, they list a couple of different theories that postulate different structural models underlying the observed variances and covariances.

10.1 Theories

Alternative theories for memory discussed are:

- Theory #1 (Berry et al.)

A single factor drives memory performance across all tests and valences.

- Theory #2 (Squire)

Two components drive memory performance, namely declarative (explicit) memory underlying associate and item memory, plus a procedural (implicit) system underlying priming.

- Theory #3 (Gardiner et al., Yonelinas)

An alternative two-component model assumes a recollection factor underlies associate memory, and a fluency factor underlies item memory and priming.

(optional variant: item memory is driven by both recollection and fluency)

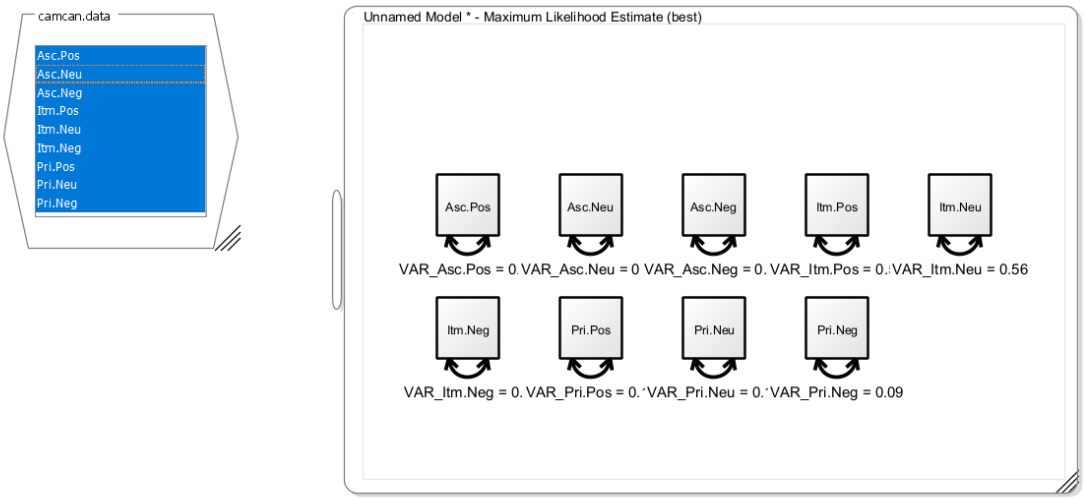
- Theory #4

A three-component model predicts that each test is best described by an individual underlying component.

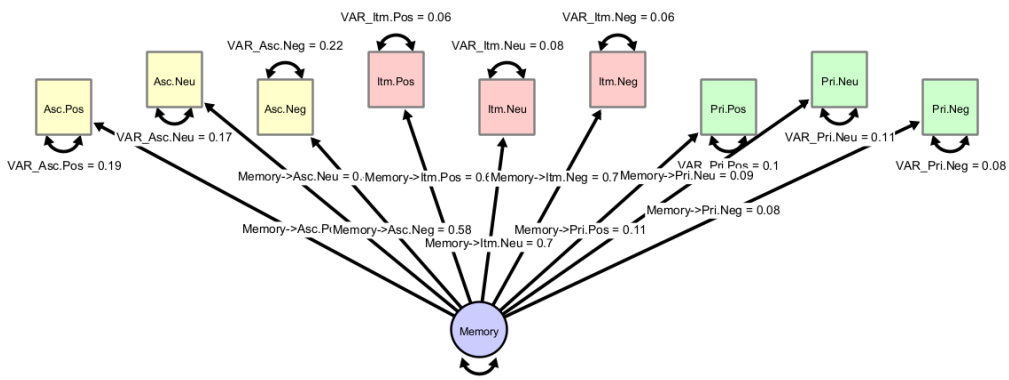
10.2 Fast Model Creation

The data are provided in the file `camcan.data`, which contains the variances and covariances across the nine indicators measured in N=305 persons.

Note that the fastest way to start modeling is to load the data, select all variables in the loaded dataset and drag them onto the empty model panel. This way, all observed variables are created in the model, together with each a free parameter representing their variance, and they are directly linked to the dataset (for model fitting). Maximum likelihood-estimation directly starts and the values represent the maximum-likelihood estimates of the variances of each variable.



In order, to create a common factor, double-click on an empty space in the model view. A latent variable (shown as a circle) appears. Change the default name to "Memory". Now add the factor loadings. These are directed paths from the latent factor to each observed variable and make them freely estimated variables. Feel free to use the layout and graph settings to customize the appearance. Note that a factor model needs an identification constraint to identify the latent scale. To implement this, either fix the latent variance to one or fix one of the loadings to one. In the first case, the latent scale is identified such that it has unit variance; in the second case, it is identified to have the same unit of measurement as the item with the unit factor loading. Your graph should look somewhat like this:



Let us inspect the explained variance in each item. You can do this visually by giving the observed variables a fill color and then select "Customize Variable -> Select Fill Style -> Bokers R2" and each observed variable will be filled according to the level of explained variance. Or, you hover the mouse over a single, unselected observed variable. A popup window will appear, in which you can read off the R^2 value as estimate of explained variance. Using either way, we can already see that the priming variables are not well explained by the common factor. Further, we find these goodness-of-fit values:

```
RMSEA (classic)                : 0.288
SRMR (covariances only)        : 0.142
CFI (to independent model)     : 0.73
```

What other models could explain these data better?

11 Exercises

Exercise:

1) Set up models representing the alternative theories, fit the models with the provided data (camcan.data) as described above and discuss the results.

2) Inspect parameter estimates (e.g., check for problems like negative variances) and model fit statistics

3) What model fits best?

4) How could your final model incorporate predictors of behavior, e.g. grey matter volume estimates of some ROIs?